

Article Methods Draft

2. Materials and Methods

2.1. Image Preprocessing and Tiling Strategy

High-resolution microscopy scans were acquired in Zeiss .czi format. Due to the gigapixel dimensions of the raw slides, which exceed the memory constraints of modern convolutional neural networks, a deterministic tiling algorithm was employed. First, a Maximum Intensity Projection (MIP) was applied to the Z-stacks to collapse the three-dimensional volumetric data into a two-dimensional representation while preserving the structural topography of the pollen grains.

The resulting macroscopic images were dynamically partitioned into overlapping 640×640 pixel tiles. Overlap was strictly maintained to prevent truncation of pollen grains intersecting the grid boundaries. To suppress false-positive detections caused by inorganic debris, air bubbles, and slide artifacts, purely empty background tiles were explicitly sampled and injected into the training dataset as “Hard Negatives” (tiles with image data but lacking bounding box labels).

2.2. Two-Stage Deep Learning Architecture

To maximize both detection recall on heterogeneous biological backgrounds (such as messy stigmas) and taxonomic classification precision, the detection pipeline was bifurcated into a two-stage cascaded architecture using the Ultralytics YOLOv11 framework.

Stage 1: General Pollen Object Detection A YOLOv11 model was trained to detect any biological structure morphologically consistent with a pollen grain. During this stage, all species-specific class labels were collapsed into a singular `pollen` class (Class ID 0). This stage prioritized high-recall boundary detection over taxonomy.

Stage 2: Species-Specific Instance Segmentation and Classification Following Stage 1 inference, the identified bounding boxes were utilized to dynamically isolate the pollen grains from the source image. These were then fed into a secondary YOLOv11-seg model. This model

was trained on pseudo-labeled reference pollen extracted directly from pristine anther scans (Pollen Production). This two-stage decoupling prevents the classification model from learning irrelevant stigma background noise.

2.3. Data Collection: Production vs. Deposition

Biological samples were systematically collected and analyzed under two distinct regimes:

1. **Pollen Production:** Samples were extracted directly from the anthers of collected specimens (e.g., *Ranunculus adoneus*, *Gentiana algida*, *Sedum lanceolatum*, *Caltha chionophila*, *Viola adunca*). These pristine samples provided the pure, mono-specific reference material necessary for training the species-specific Stage 2 classification model and measuring absolute pollen volume per flower.
2. **Pollen Deposition:** Samples were collected from stigmas or natural environmental traps to evaluate inter-species pollen transfer and pollination network dynamics. The dual-stage model independently predicted the taxonomic identity of each pollen grain deposited on these surfaces, allowing for quantitative time-series analysis of conspecific versus heterospecific pollen loads.

2.4. Continuous Active Learning and Annotation

To continuously adapt the model to novel pollen orientations and previously unseen biological debris, a human-in-the-loop Active Learning interface was developed. The system allows domain experts to rapidly triage Stage 1 predictions on novel data: 1. **Acceptance:** Perfect segmentations are committed immediately to the `Curated_Retrain_Data` repository. 2. **Hard Negatives:** Erroneous detections (e.g., misidentified debris) are rejected and stored with empty annotation matrices to explicitly penalize the model in future training cycles. 3. **Manual Refinement:** Imprecise polygons are passed through a data pipeline to Roboflow, where experts manually redraw the physical boundaries using semantic polygon tools before reintegrating them into the training corpus.

2.5. Computational Environment and Inference

Model training and heavy preprocessing workloads were orchestrated as distributed containerized jobs via Kubernetes on the CESNET MetaCentrum High-Performance Computing cluster, utilizing NVIDIA A100 and A40 Tensor Core GPUs. Post-inference, the pipeline reconstructs the 640×640 predictions into downscaled 2000-pixel macro-stitches with superimposed polygon masks for rapid visual verification, alongside tabular CSV exports containing spatial coordinates and morphological area (μm^2) metrics for each detected grain.